

Homework 2

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Question 1

```
install.packages("aRtsy")
```

`install.packages` installs the package and all of its content onto the computer. On the other-hand, `library` loads an installed package into the current document or the code chunk, making it easier to access.

Packages

Question 2

```
library(aRtsy)
```

The collatz conjecture is an unsolved math problem that asks if any positive number can be brought down to 1 by repeating only two rules

n = chosen number

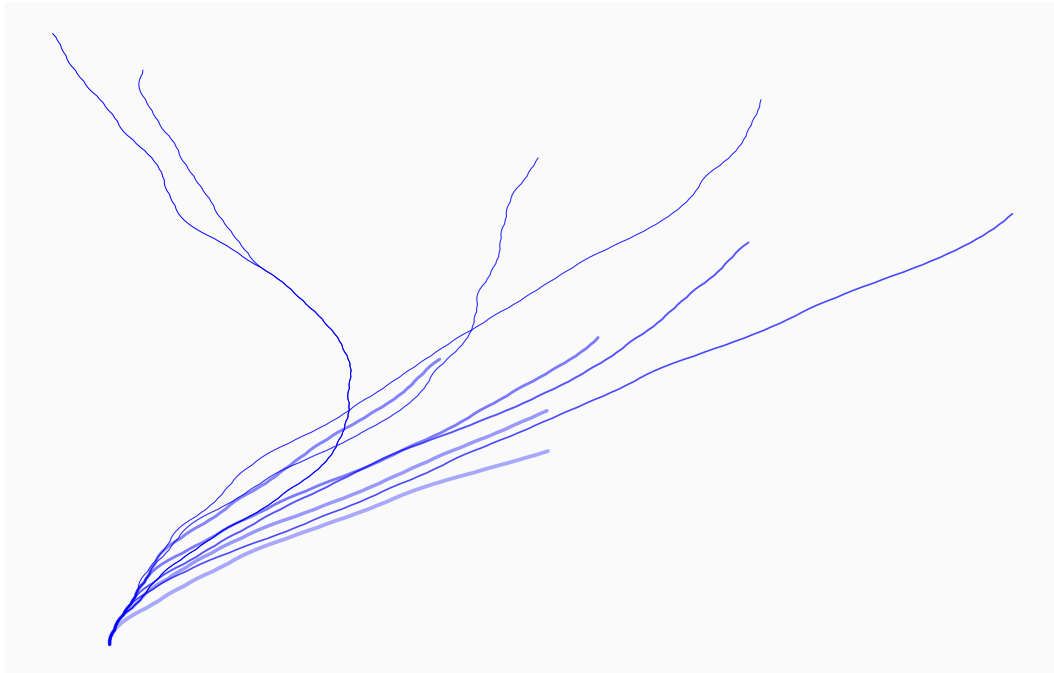
if n is even, $n/2$

if n is odd, $3n + 1$

Question 3

a

```
canvas_collatz(  
    colors = c("blue"),  
    n = 10  
)
```



The `canvas_collatz` function draws the collatz conjecture on a canvas, taking into account the arguments. In the code chunk above, the argument ‘color’ specifies what color will be used when the function maps the conjecture, in this case, blue. In addition, the ‘n’ argument is the positive integer that determines how many random starting integers will be used to draw the lines, which is 10 for the code chunk above.

b

There is a different output every time the code is ran because only the number of random integers can be changed. Therefore the integers itself are still random, meaning there will be a very unlikely chance of generating two identical images

This might be troublesome when trying to put the code in the qmd file because every output is different, meaning that whichever image you are attempting to display will come out differently when rendered again.

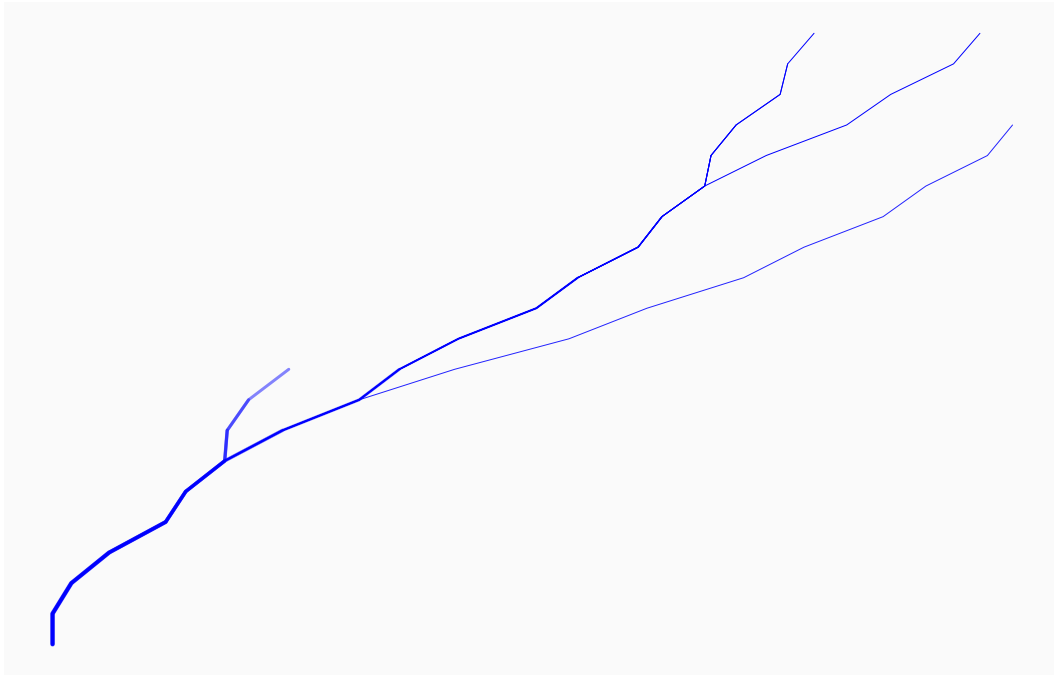
The `set.seed` function makes the random number generator the same every time since it gives it the same starting value every time

```
set.seed(8)
canvas_collatz(
  colors = c("blue"),
  n = 10
)
```



Question 4

```
canvas_collatz(  
  colors = c("blue"),  
  n = 1:20  
)
```



The main difference in the code here and the one in the previous chunk is that this code does not need a `set.seed` function to generate the same image every time. However, instead of `n` being a single number, `n` in this code chunk is every number from 1 to 20, meaning that it is not choosing 10 random integers, rather the chosen integers are 1 to 20 and are already set